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SYSTEMS AND METHODS FOR HARNESSING MARINE HYDROKINETIC ENERGY

Abstract

An example system comprises an enclosure submerged in a body of water. The system also comprises an intake port disposed along a periphery of the enclosure to transport water into the enclosure. The system also includes a turbine generator disposed inside the enclosure and coupled to the intake port to receive the water entering the enclosure through the intake port. The system also comprises a water storage tank coupled to the turbine generator to receive the water flowing out of the turbine generator. The system also comprises a pump coupled to the water storage tank to pump the water out of the water storage tank. The system also comprises a controller to control flow of the water into the enclosure by operating the intake port and to control flow of the water out of the enclosure by operating the pump.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/020,855 filed on Feb. 10, 2023, which is a national stage entry of International App. Serial No. PCT/US2021/045605 filed on Aug. 11, 2021, which claims priority from U.S. Prov. App. No. 63/064,769 filed on Aug. 12, 2020 and U.S. Prov. App. No. 63/179,897 filed on Apr. 26, 2021, the entirety of each of which are incorporated herein by reference.

FIELD

[0002] This invention generally relates to power systems and more particularly to systems and methods for harnessing marine hydrokinetic energy.

BACKGROUND

[0003] Power generation and energy storage technologies are continuously evolving to accommodate our ever increasing demand for electricity.

[0004] Some power systems rely on non-renewable energy sources, such as coal, natural gas, nuclear fuel, petro-chemicals, and other fuels, to generate a predictable amount of power when needed by increasing or decreasing the amount of fuel input into the system according to energy demand at any given time. However, such power systems may also be associated with significant or unpredictable financial costs (e.g., due to fuel market price fluctuations, fuel extraction and mining costs, etc.) as well as environmental costs (e.g., carbon emissions, damage caused by mining operations, etc.).

[0005] Some power systems mitigate or avoid these disadvantages by instead relying on renewable energy sources, such as solar energy (e.g., solar cell systems), wind energy (e.g., wind turbine systems), etc., which are generally more abundant and can usually be harvested without causing as much damage to the environment compared to fuel sources. However, there are also other technical challenges associated with harnessing renewable energy efficiently. For example, some renewable energy systems passively collect energy at less predictable times and/or rates depending on the current state of their surrounding environment. For instance, power output by a solar power system (e.g., photovoltaic cells) or a wind power system at any given time will vary depending on the current weather conditions (e.g., cloudy vs. clear sky, local wind speeds, etc.) in their surrounding environment.

[0006] Hydrokinetic energy is another type of renewable energy source, and is generally the energy that drives the movement of bodies of water. Tides, waves, ocean currents, and free-flowing rivers contain vast amounts of largely untapped, powerful, and clean hydrokinetic energy. Natural bodies of water can store immense amounts of hydrokinetic energy over time due to the thermal energy of the sun's heat and the mechanical energy exerted by the gravitational pull of the moon and the sun.

[0007] Traditional hydropower systems can sometimes be used to generate electricity efficiently (when needed). For example, a river can be dammed to accumulate the hydrokinetic energy of its flowing water by filling a reservoir behind the dam to convert it into potential energy. The stored water can then be released from the reservoir selectively (when needed) through a water turbine in a controllable manner (e.g., by controlling the rate of water flow through the water turbine) to generate an electricity signal having desired characteristics. However, hydropower dams may be less suitable or practical in some geographic locations (e.g., where there is no nearby river, or when the surrounding terrain is not ideal for damming a deep reservoir). Additionally, damming a river

may sometimes result in other types of environmental harm, safety risks (e.g., flooding, etc.), and/or interfere with the use of the adjacent lands. Moreover, such traditional inland hydropower systems may not be suitable for harnessing the even more massive amounts of hydrokinetic energy stored in other bodies of water, such as oceans and seas.

[0008] The present disclosure is directed to power systems that provide significant advantages and capabilities over prior art power systems of the type discussed above.

SUMMARY

[0009] The present disclosure provides systems, methods, and apparatus that enable producing and/or storing electrical power efficiently by way of harnessing hydrokinetic energy at high pressure underwater environments.

[0010] In an example, a system is provided that includes an enclosure submerged in a body of water. The system also includes an intake port disposed along a periphery of the enclosure to transport water into the enclosure. The system also includes a turbine generator disposed inside the enclosure and coupled to the intake port to receive the water entering the enclosure through the intake port. The system also includes a water storage tank coupled to the turbine generator to receive the water flowing out of the turbine generator. The system also includes a pump coupled to the water storage tank to pump the water out of the water storage tank. The system also includes a controller to control flow of the water into the enclosure by operating the intake port and to control flow of the water out of the enclosure by operating the pump.

[0011] In another example, a system is provided that includes an enclosure submerged in a body of water. The system also includes an intake port disposed along a periphery of the enclosure to transport water into the enclosure. The system also includes a turbine generator disposed inside the enclosure and coupled to the intake port to receive the water entering the enclosure through the intake port. The system also includes a discharge port disposed along the periphery of the enclosure to transport the water exiting the turbine generator out of the enclosure. The system also includes a controller to control flow of the water into the enclosure by operating the intake port.

[0012] Additional features and advantages of the disclosed systems, apparatus, and methods are described in, and will be apparent from, the following Detailed Description and the Figures. The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the figures and description. Also, any particular embodiment does not have to have all of the advantages listed herein. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes, and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0013] FIG. 1 is a block diagram of an example power system, according to an example embodiment of the present disclosure.

[0014] FIG. 2 illustrates an alternate embodiment of an example power system, according to the present disclosure.

[0015] FIG. 3 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0016] FIG. 4 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0017] FIG. 5 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0018] FIG. 6 illustrates another alternate embodiment of an example power system, according to

the present disclosure.

[0019] FIG. 7 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0020] FIG. 8 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0021] FIG. 9 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0022] FIG. 10 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0023] FIG. 11 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0024] FIG. 12 illustrates another alternate embodiment of an example power system, according to the present disclosure.

[0025] FIG. 13 illustrates another alternate embodiment of an example power system, according to the present disclosure.

DETAILED DESCRIPTION

[0026] Oceans and seas have vast amounts of untapped hydrokinetic energy, including tidal energy sufficient to drive massive tidal waves across our planet every day. Oceans also store huge amounts of gravitational potential energy as evidenced by the immense pressures in deep ocean waters. For example, water pressures at the average ocean depth of about 12,100 feet can be approximately 5,239 pounds per square inch (PSI). As another example, some oil industry rigs deployed today include: Auger Platform, seabed depth 2,720 feet, ocean pressure 1,177 PSI; Mars Platform, seabed depth 2,940 feet, ocean pressure 1,273 PSI; Horn Mountain Platform, seabed depth 5,422 feet, ocean pressure 2,347 PSI; Perdido Platform, seabed depth 8,040 feet, ocean pressure 3,481 PSI. In general, seabed or ocean floor underwater pressures at many different locations are suitable for various applications of the example power systems of the present disclosure.

[0027] The present disclosure includes example systems and methods for harnessing marine hydrokinetic energy to enable or improve various power applications (e.g., power generation, power conversion, energy storage, etc.). The systems and methods described herein are not limited to harnessing energy at ocean depths, but can also include harnessing energy at depths of other bodies of water such as, but not limited to, rivers, seas, lakes, etc.

[0028] FIG. 1 is a block diagram of an example embodiment of a power system **100**, according to the present disclosure. In the illustrated example, the system **100** includes a submersible enclosure **110**, a tube **118**, an air compressor **119**, a water storage tank **120**, a plurality of sensors **130**, and a controller **140**.

[0029] The enclosure **110** defines a sealed low pressure environment (e.g., air, vacuum, etc.) configured to be submerged under water at various depths such that water pressure outside the enclosure **110** is substantially greater than pressure inside the enclosure. In some examples, the enclosure **110** has a spherical shape, a cylindrical shape, or any other shape suitable to withstand the pressure differential between the water surrounding the enclosure **110** (e.g., when submerged underwater) and the low pressure environment inside the enclosure **110**.

[0030] In some examples, the enclosure **110** can be a main structure of the system **100**. For example, the enclosure **110** can be configured as an underwater power plant. In some examples, enclosure **110** is mounted to a seabed. In other examples, the enclosure **110** is deployed at different depths between the seabed and the surface of the body of water in which the enclosure **110** is submerged. In some examples, the enclosure **110** may include compressed air storage tanks, water storage tanks, or any other component required to operate the enclosure **110** as a submarine, such as ballast tank(s), variable ballast tank(s), trim tank(s), vent valve(s), a pressure hull and/or any other component suitable for operating the enclosure **110** as a remotely piloted submarine. For example, the enclosure **110** can be operated remotely (e.g., using the controller **140**) to move the enclosure

upwards or downwards to different sea levels or depths.

[0031] In some examples, the enclosure **110** can include one or more compartments, and each compartment can be pressurized or non-pressurized.

[0032] In some examples, the system **100** includes more than one enclosure **110** submerged under the same body of water and arranged at different positions relative to one another.

[0033] In the illustrated example, the enclosure **110** includes a plurality of ports **112**, a turbine generator **114**, and a piping system/mechanism **116**.

[0034] The plurality of ports **112** are disposed along a periphery of the enclosure **110** to define channels through which water can flow into or out of the enclosure **110**. To that end, in some examples, a port **112** includes a valve **112** (e.g., pressure valve, a vent valve, etc.), a nozzle **112**, a waterjet **112**, combinations thereof, and/or any other component that is operable to allow, prevent, or adjust a flow of water into or out of the enclosure **110** at the port. For example, a valve **112** can be used to increase or decrease the flow of water transported into or out of the enclosure **110** by at least partially closing or opening the valve. In some examples, one or more ports **112** are intake port(s) **112** or inlet(s) **112** that transport water, compressed air, or any other fluid into the enclosure **110**. In some examples, one or more ports **112** are discharge port(s) **112** or outlet(s) **112** that transport water, air, or any other fluid out of the enclosure **110**.

[0035] In some examples, the plurality of ports **112** are arranged at different positions along the periphery of the enclosure **110** and/or oriented in a plurality of directions relative to a body of water surrounding the enclosure **110**. For example, where the enclosure **110** has a spherical shape, each port **112** may transport water from a different direction relative to the body of water depending on the position of the port on the enclosure **110**.

[0036] Thus, in some examples, the ports **112** include one or more intake valves, outlet valves, or a combination thereof. In some examples, the ports **112** include water intake valves, water outlet valves, air intake valves, air outlet valves, or a combination thereof. The valves can be Penstock Pipe or composed of a Penstock type pipe. In other embodiments, on/off stop valves **112** can be placed near intake valves, outlet valves, or a combination thereof.

[0037] The turbine generator **114** is disposed inside the disclosure (e.g., in the low pressure environment) and coupled to an intake port **112** to receive water entering the enclosure **110** through the intake port. The received water may flow through the turbine to cause the turbine to generate power (e.g., by rotating turbine blades, etc.). The turbine generator **114** is also coupled to the water storage tank **120**. Thus, the system **100** directs water flowing out of the turbine generator **114** into the water storage tank **120**. In this way, the generator **114** can continue to generate power based on the pressure differential between the high pressure water outside the enclosure **110** and the low pressure environment inside the enclosure **110** (e.g., by using the water storage tank **120** to hold the spent water flowing out of the generator **114**).

[0038] Turbine generator **114** may include any type of turbine generator, such as a Francis turbine, a Pelton turbine, a Kaplan turbine, a Deriaz turbine, a Jonval turbine, a Cross turbine, a reaction type turbine, an impulse type turbine, or any other type of turbine. The turbine generator **114** can be mounted vertically, horizontally, or in any suitable orientation depending on an application of the system **100**.

[0039] The piping system **116** is disposed inside the enclosure **110** and configured to couple an inlet of the turbine generator **114** to an intake port of the ports **112** and/or to couple an outlet of the turbine generator **114** with the water storage tank **120**. To that end, in some examples, the piping system or mechanism **116** includes one or more pipes (e.g., internal pipes) disposed inside the enclosure **110**. As an example, the piping system **116** may include pipes connected to intake valve(s) or nozzle(s) or port(s) **112**, which may be penstock pipes or other types of pipes capable of transporting high pressure water from the intake valve(s) or nozzle(s) or port(s) **112** to the turbine **114**. Additionally or alternatively, as another example, the piping system **116** may include pipe(s) connected to the turbine **114** to transport spent water flowing out of the turbine generator **114** to the

water storage tank **120** and/or pipes to transport the spent water out of the enclosure **110** via a discharge port(s) or outlet(s) or valve(s) **112**. Additionally or alternatively, in some examples, the system **100** includes one or more external pipes (not shown) disposed outside the enclosure **110**. For example, the external pipes may transport water between the enclosure **110** and another enclosure (not shown). As another example, the external pipes may transport water between the enclosure **110** and the water storage tank **120**.

[0040] The tube **118** may include any type of tube connected to a port **112** and formed from a structure that can transport high pressure fluids (e.g., high pressure water, compressed air, etc.) into or out of the enclosure **110** via the port **112**. As an example, the tube **118** may be coupled to a discharge port or valve **112** and may extend away from the discharge port **112** to define a channel for transporting water (flowing out of the enclosure **110** at the discharge port **112**) to the water storage tank **120**. As another example, the tube **118** may alternatively include a tube that transports high pressure air (or compressed air) into an intake port **112** of the enclosure **110**. As another example, the tube **118** may alternatively be arranged to transport water from a discharge port **112** of the enclosure **110** to a surface of the body of water in which the enclosure **110** is submerged. In another example, the tube **118** can alternatively be arranged to transport water from the water storage tank **120** to the surface of the body of water.

[0041] The air compressor **119** includes any type of air compressor. In some examples, the air compressor **119** is configured to deliver compressed air (or high pressure air) into the enclosure **110** and/or the water storage tank **120**. In an example, the air compressor **119** is coupled to an intake port **112** to provide compressed air entering the enclosure. To that end, the air compressor **119** can be submerged inside the body of water (e.g., near the intake port) or disposed outside the body of water (e.g., on a surface vessel or an onshore platform).

[0042] Alternatively or additionally, in some examples, the air compressor **119** can be employed as an air vent **222** that pumps air out of the enclosure **110** and/or the water storage tank **120**.

[0043] In some examples, the air compressor **119** can provide higher pressure air even at high pressure ocean depths. For example, the air compressor **119** can be employed to pump water exiting the enclosure **110** at a discharge port **112** toward the water storage tank **120** and/or toward a surface vessel or onshore platform.

[0044] The water storage tank **120** is coupled to the turbine generator **114** to receive (and store) the water flowing out of the turbine generator **114** (e.g., spent water).

[0045] In some examples, the water storage tank **120** is disposed outside the enclosure **110**. For example, the water storage tank **120** may be coupled to a discharge port **112** of the enclosure **110** via the tube **118** to receive water flowing out of the enclosure **110** through the discharge port **112**. Thus, in this example, the tube **118** may define a channel or passageway for transporting the discharged water from the enclosure **110** to the water storage tank **120**. In other words, the water storage tank **120** can be connected/attached to the enclosure **110**. In other examples, the water storage tank **120** is disposed inside the enclosure **110**. For example, the water storage tank **120** may be a component of the enclosure **110**. In other examples, the enclosure **110** and the water storage tank **120** are a single structure. For example, the enclosure **110** can store the water exiting the turbine generator **114** inside the enclosure **112** instead of discharging it out of the enclosure **110** to the water storage tank **120**.

[0046] In some examples, the storage tank **120** receives water from the enclosure **110** (via the tube **118**) at a relatively lower flow rate or gallons per minute (GPM), and discharges water out of the storage tank **120** at a relatively higher flow rate or GPM rate. For example, the storage tank **120** can be configured to discharge water at a same or faster rate than the water flowing into the storage tank **120** to avoid reaching a full capacity. The storage tank **120** can be pressurized or non-pressurized.

[0047] In the illustrated example, the water storage tank **120** includes a water pump **122**, a buoyancy structure **124**, and an air pump **126**.

[0048] The water pump **122** is configured to pump water out of the water storage tank **120** (e.g., back to the body of water in which the enclosure **110** and/or the storage tank **120** is submerged). [0049] The water pump **122** can be disposed inside or outside the water storage tank **120**. In an example, the water pump **122** is located on a vessel floating on the body of water in which the water storage tank **120** is submerged. More generally, the water pump **122** can be used to discharge water from the water storage tank **120** or from the enclosure **110** (e.g., via outlet valves or ports **112**). In other embodiments, the water pump **122** can discharge water toward an ocean or ship-based turbine generator **114**.

[0050] The buoyancy structure **124** may include ballast tanks, balloons, and/or any other structure disposed inside or coupled outside the water storage tank **120**. The buoyancy structure **124** is configured to receive air (pumped by the air pump **126**) to increase buoyancy forces exerted on the water storage tank **120**. For example, air pump **126** may pump the air into a ballast tank **124** to reduce a density of the water storage tank **120** relative to the density of surrounding water. In this way, the water storage tank **120** may rise to a lower ocean depth where discharging water stored in the storage tank **120** does not require a same amount of energy as when the water storage tank was at a greater ocean depth. Thus, the air pump **126** can include any type of air pump suitable for pumping air into the buoyance structure.

[0051] The sensors **130** include pressure sensors and/or underwater current sensors. In some examples, the sensors **130** include a pressure sensor disposed at or near the water storage tank **120** to provide measurements of water pressure outside the water storage tank **120**. For example, the pump **122** can be operated (e.g., via the controller **140**) to adjust the flow of water pumped out of the water storage tank **120** depending on the measurements by the pressure sensor **130**. For example, if the pressure is lower than a threshold pressure (e.g., due to the water storage tank **120** moving up to a threshold depth level from a surface of the body of water, due to a low tide event at the location of the storage tank **120** when the pressure sensor measured the pressure, etc.).

[0052] In some examples, the sensors **130** include an underwater current sensor or other type of sensor that provides measurements indicative of a direction of an underwater current at or near a location of the enclosure **110**. In these examples, the system **100** can be configured to select an intake port **112** (e.g., oriented opposite to a direction of the underwater current), and then route water from the selected intake port **112** (e.g., by operating the piping mechanism **116**) into the turbine generator **114**. In this way, the system **100** advantageously receives water having more hydrokinetic energy because the selected intake port **112** will be receiving the high pressure water in alignment with a natural direction of water flow in the underwater current.

[0053] The controller **140** is configured to regulate and/or control the flow of water into and/or out of the enclosure **110** and/or the storage tank **120**. To that end, the controller **140** may include a computer that has hardware and/or software executable to control the system **100** in accordance with the present disclosure. For example, the computer **140** may include one or more processors and a memory device (e.g., a non-transitory computer readable medium) storing instructions that, when executed by the one or more processors, cause the computer **140** to perform the functions described herein. Alternatively or additionally, the controller **140** may include digital and/or analog circuitry wired to perform the functions of the controller **140** described herein.

[0054] The controller **140** may be configured to control flow of water into the enclosure **110** by operating an intake port **112** of the enclosure **110** (e.g., by switching an intake port or valve **112** to allow, prevent, or adjust the rate at which the water is flowing in). Similarly, the controller **140** may also be configured to control flow of water out of the enclosure **110** and/or the water storage tank **120**.

[0055] In an example, the controller **140** increases a buoyancy force exerted on the water storage tank **120** by operating the air pump **126** to pump air into the buoyancy structure **124**. By increasing the buoyancy force, the controller **140** in this example may thus cause the water storage tank **120** to move upwards toward a surface of the body of water in which the water storage tank **120** is

submerged. In this example, the controller **140** can optionally delay or slow down the rate at which water is being pumped out of the water storage tank **120** until the water storage tank **120** reaches a surface of the body of water or at least a threshold depth level (e.g., where water pressure is relatively lower) before starting or increasing the rate at which water is being pumped out of the water storage tank **120**. In this way, energy needed to pump the spent water out of the water storage tank **120** can be reduced.

[0056] In an example, the controller **140** is configured to adjust flow of the water pumped out of the water storage tank **120** (and/or into the enclosure **110**) based on a tidal phase of the body of water (in which the tank **120** is submerged) at a location of the water storage tank **120** and/or the enclosure **110**.

[0057] In general, tidal waves have very long wavelengths (e.g., miles or tens of miles) such that, at any particular location in the ocean, high tides and low tides occur in a predictable manner (e.g., approximately every 6 hours, 12 hours, etc.). Thus, the controller **140** can start pumping and/or increase the rate at which water is being pumped out of the storage tank **120** during times when the tidal phase is at or near a low tide. Similarly, the controller **140** can start or increase the rate at which water is being transported into the enclosure **110** when the tidal phase is at or within a threshold from a high tide (e.g., when the sea level above the enclosure **110** is slightly higher than average due to the tidal forces exerted on the oceans by the moon and/or the sun).

[0058] Thus, the example systems herein advantageously enable harnessing tidal energy by selectively allowing water into the enclosure **110** and/or the storage tank **120** during high tides (when the water level is relatively higher), and/or by delaying the timing or rate of pumping water out of the water storage tank **120** until a low tide condition occurs (i.e., when the water level above the storage tank **120** is relatively lower) to harness the pressure differential between the high tide and the low tide.

[0059] Notably, the example systems herein also advantageously improves the efficiency of harvesting tidal wave energy, by using the high pressure environment at the ocean depths to more effectively transport large amounts of water through the turbine **114** during the high tide phase. For example, if the turbine was instead closer to the surface of the body of water, a relatively smaller amount of water will flow through the turbine **114** during the high tide phase. Accordingly, the example systems herein provide a significant advantage over traditional systems by efficiently harnessing one of the relatively more challenging but very abundant sources of marine hydrokinetic energy.

[0060] In some embodiments, the system **100** can be deployed at ocean depths far above a seabed or on the seabed. Options can be tethered to power distribution center(s), electrical grid(s), ocean vessel(s) and or rig(s), ship(s), boat(s), yachts(s), vehicle(s) in order to deliver electrical energy.

[0061] In some embodiments, the enclosure **110** can be configured as an ocean vessel including a propulsion system for navigation and positioning the enclosure **110** at sea depths. In some embodiments, the enclosure **110** can employ submarine components required to remotely pilot the enclosure **110** to a suitable location. The suitable location, for example, can include, but is not limited to, land, or another part of the ocean.

[0062] In other embodiments, the enclosure **110** can be deployed inside a vessel as an electric energy power source for the vessel. In some embodiments, the vessel can be a submarine.

[0063] In some embodiments, the system **100** can include a substance (e.g., inside the enclosure **110** or inserted into the enclosure **110** at an intake port **112**) to increase the pressure of water flowing through and rotating the turbine generator **114**. The substance can be a liquid, a gas, an additive, plasma, the like, or a combination thereof.

[0064] In some embodiments, the system **100** includes cables. The cables can be submarine cables. In some embodiments, cables and submarine cables can be employed. The cables can deliver the electrical energy to the industry. It can be electrical power, and signal cables for operation.

[0065] In some embodiments, the system **100** can include one or more vessels. Vessels can include,

but are not limited to, a rig, a ship, a yacht, a submarine, a vehicle, a boat, a combination thereof, or any type of ocean vessel.

[0066] In some embodiments, the system **100** includes a transmission gear box to increase the production of electricity. Transmission gear boxes utilized in the renewable energy industry can be employed.

[0067] In some embodiments, the system **100** can include transformer(s), control room(s), computer(s) **140**, and/or battery storage unit(s), or a combination thereof. In other embodiments, the systems can include transformer(s), control room(s), robotic(s), automation, computer(s) **140**, and/or battery storage unit(s), or a combination thereof, whether within the system **100**, outside the system **100**, or near the system **100**, or in the sea depths, or placed on land. In other embodiments, the system **100** can include accommodations for human habitation for periods of time, such as oxygen supply, sleeping quarters, kitchen, bathrooms, showers, storage area, living quarters, and is not limited to the list herein can be interchangeable and employed in the system **100**.

[0068] In some embodiments, the system **100** can include a power distribution center configured to process raw electricity to be distributed to an electrical grid. The electrical grid can power one or more cities.

[0069] It should be appreciated that the system **100** can alternatively include fewer or more components than those shown, and/or one or more of the components shown can be alternatively implemented as a single structure that performs the combined functions of the one or more components.

[0070] For example, in various embodiments, the system **100** can comprise one or more enclosures **110**, one or more vessels, one or more power distribution centers, one or more storage tanks **120**, one or more intake valves **112**, one or more outlet valves **112**, one or more pressure valves, one or more turbine generators **114**, one or more transmission gears, one or more water pumps **122**, one or more computers **140**, one or more motors, one or more tubes **118**, electronics, a mount, and/or one or more air compressors **119**. In some embodiments, the system **100** includes hardware and/or software commonly used in marine hydrokinetic electrical energy systems.

[0071] FIG. 2 illustrates an alternate embodiment **200** of an example power system, according to the present disclosure.

[0072] In some embodiments, the enclosure **110** can have any shape, such as the shape shown in FIG. 2 or another shape, that concludes as a sealed device configured to prevent high pressure water and/or air from escaping, except through intake valves **112a**, **112b** or outlet valve **112c** (collectively referred to as ports **112**) that control high pressure water and/or air flow into or out of the enclosure **110**. In other embodiments, the enclosure **110** can have one or more main compartments and options for smaller compartments. In the illustrated example of FIG. 2, the enclosure **110** includes two main compartments **210a**, **210b**. The main compartment **210b** can include the turbine generator **114**, valve(s), and/or gear boxes(s), power transmission, computer, and/or other electronics required to operate the system **200**. In this example, the other main compartment **210a** can include the high pressure water flow used to rotate blades of the turbine generator **114**.

[0073] The enclosure **110** can include one or more turbine generators, one or more valves and/or one or more gear boxes, and a power transmission and/or electronics required to operate the system **200**. Outside the enclosure **110**, covering the intake **112a** is an optional baffle to prevent wildlife (e.g., fish) from blocking the intake **112a**. The turbine generator(s) **114**, valve(s), gear box(es), power transmission, computer, and or electronics can be placed in a controlled pressurized compartment **110b** within the enclosure **110**. The turbine generator blades can be exposed to high pressure water flow for rotation inside the enclosure **110**. In some embodiments, the system **200** includes a tube **220** supplying high air pressure from vessel **230** to the enclosure **110**. The air supply can be used to operate the enclosure **110** and regulate the proper pressure in the water flow compartment **210b** of the enclosure **110**. In other embodiments, the enclosure **110** can be deployed

at 1,000 feet ocean depth and the PSI can be 433, a higher PSI can be necessary to get the flow of water to the sea surface. Thus, added air pressure of PSI **200** to reach PSI **633**, which is greater at the deployed sea level depth of 1,000 feet can be employed.

[0074] In other embodiments, high pressure water can enter the enclosure **110** through an intake valve **112a** and rotate the turbine generator **114** to produce electricity. The electricity can then be sent to the vessel **230** for distribution to an electrical grid **240** by way of one or more electrical cables **250**.

[0075] In some embodiments, the enclosure **110** can employ one or more stop valves **260** to be used as emergency turn on/off lever(s) controlling high pressure water flow. The electricity can be produced by hydrokinetic energy obtained from water pressure at ocean depths, rotating the turbine generator **114**.

[0076] Once high pressure water exits the enclosure **110** through the exit valve **112c**, it can enter the tube and/or pipe **118** that can deliver high pressure water to the water storage tank **120** on its way to vessel **230**. High pressure water can be extracted from the water storage tank **120** using a water pump **122**. Exiting high pressure water can be routed to the vessel **230** and/or connected to an on board turbine generator or water pump.

[0077] In some embodiments, the enclosure **110** can be mounted to the seabed (ocean floor) and/or tethered to a vessel **230** at the ocean's surface.

[0078] Enclosures **110** can be deployed in ocean(s), sea(s), river(s), lake(s), manmade lakes(s) and any body of water. In some embodiments, sphere enclosures can operate in water depths and land holes filled with water at depths. In other embodiments, water pressure at ocean depths can be the main ingredient required to produce hydrokinetic electrical energy.

[0079] The enclosure **110** can be composed of any suitable material used in the hydrokinetic energy industry. Water pressure, other liquids pressure, and/or pressurized air can be used for operations in the system **200**.

[0080] In some embodiments, the enclosure **110** can employ one or more motors with thrust propulsion used to maneuver the enclosure **110** into a proper position to operate at sea depths.

[0081] The enclosure **110** can employ one or more air tubes **220** from the enclosure **110** to the vessel **230**. The enclosure **110** can be designed to supply air pressure when required for operations and maintenance.

[0082] The enclosure **110**, in the illustrated example of FIG. 2, can produce electricity by way of hydrokinetics energy, obtained from water pressure at ocean depths. In some embodiments, the system **200** comprises the enclosure **110** with intake **112a** for high pressure water to enter the enclosure **110**. The flow of high pressure water can rotate turbine generator(s) **114**, which can produce electricity and send it to the water storage tank **120** then to the vessel **230**. The vessel **230** can include, but is not limited to, an ocean surface vessel, a rig, a ship, a boat, a yacht, a vehicle, land, or a building. Then from the vessel **230** to a power distribution center **250**. In some embodiments, the power distribution center **250** can be on land. In other embodiments, the enclosure **110** can produce electricity from water pressure at ocean depths. In some examples, the system **200** includes, but is not limited to, the enclosure **110**, the turbine generator(s) **114**, valve(s) **112**, transmission gear(s), water pump(s) **122**, air compressor(s) **119**, storage tank(s) **120**, ocean vessel(s) **230**, computer(s) **140**, motor(s) tube(s), cables, electronics, mounting apparatus, and tethering the unit and sensor(s).

[0083] The system **200** can be designed to produce electricity offshore. For example, the vessel **230** can be alternatively configured as a long term ship/rig platform. One or many may be deployed as a part of an electricity producing farm at sea. Operation staff employees can work and live aboard the ship or rig **230**. The ship/rig **230** can be deployed miles offshore. It can be connected to a power distribution center **250** on land and or designed to be a re-fueling, electric energy station for ocean going electric ships and vessels. The system **200** can be deployed along the world's coast and along ocean shipping routes as a re-fueling, electric energy station for ocean vessels, ships, aircraft and

drones.

[0084] FIG. 3 illustrates another alternate embodiment **300** of an example power system, according to the present disclosure. The system **300** comprises a sphere enclosure **110** or any shape enclosure **110** that concludes as a sealed embodiment (device) not to allow high pressure water and or air to escape, except through intake(s) **112a**, **112b** or outlet(s) **112c** or valve(s) that control high pressure water and/or air flow. The sphere enclosure **110** can include one or more main compartments and options for smaller compartments. One compartment can be used to store electronics required for operation. The other compartment can be used for high PSI pressure water flow used to rotate turbine generator blades, which can be located on the vessel **230** at sea surface.

[0085] The sphere enclosure **110** can include one or more valves **260** and/or electronics required to operate system **300**. Outside the sphere enclosure **110** covering the intake **112a** can be optional baffle to prevent wildlife, such as but not limited to, fish, from blocking the intake **112a**.

Electronics can be placed in a controlled pressurized compartment within the sphere enclosure **110**.

[0086] An optional tube **220** can be included to move high pressure air from the vessel **230** to the sphere enclosure **110**. The air supply can be used to operate the system **300** and regulate the proper PSI pressure in the water flow compartment of the sphere enclosure **110**.

[0087] High pressure water can enter the sphere enclosure through intake valve(s) **112a** and exit on its way to a turbine generator **114** located on vessel **230**, at the sea's surface, for distribution to the electrical grid **240** by way of one or more electrical cable(s) **250**.

[0088] The sphere enclosure **110** can employ one or more stop valves **260** to be used as emergency turn on off lever(s) to control high pressure water flow. Electricity can be produced by connecting high pressure water flow tube(s) **118** to a turbine generator **114** to produce electricity located on the vessel **230**. Hydrokinetic energy can be obtained, at least partially, from water pressure at ocean depths.

[0089] The enclosure **110** can be mounted to the seabed (ocean floor) and/or tethered to a vessel **230** at the surface.

[0090] The system **300** can be deployed in ocean(s), sea(s), river(s), lake(s), manmade lakes(s) and/or any body of water. It can operate in water depths and land holes filled with water at depths.

[0091] The system **300** component material can be composed of any suitable material used in the hydrokinetic energy industry. Water PSI, liquids PSI, and/or pressurized air PSI can be used for operations of the embodiments.

[0092] The system **300** can employ one or more motors with thrust propulsion used to maneuver the enclosure **110** into proper position to operate at sea depths.

[0093] The system **300** can employ one or more air tube(s) **220** connected from the enclosure **110** to the vessel **230** at the water's surface. It can be designed to supply compressed air when required for operations and maintenance.

[0094] The system **300**, as illustrated in FIG. 3, includes a sphere enclosure **110** with intake(s) **112a** for high pressure water to enter at high pressure. The flow of high pressure water can pass through valve(s) **112c**, exiting the sphere enclosure **110**, through a tube **118** on its way to a turbine generator **114** located on vessel **230** at the surface. Components of the system **300** include, but are not limited to, a sphere enclosure **110**, turbine generator(s) **114**, transmission gear(s), valve(s) **112**, air compressor(s) **119**, ocean vessel(s) **230**, computer(s) **140**, motor(s), tube(s) **118**, **220**, cables **250**, electronics, mounting apparatus **310**, and tethering the unit and sensor(s).

[0095] The system **300** can be designed to produce electricity offshore utilizing a mobile short-term ship/rig platform **230**. Operation staff employees can work and live aboard the ship or rig **230**. The ship or rig can be deployed anywhere near the shore in deep waters. In some examples, several ships/rigs can be joined as an electricity producing farm at sea. The turbine **114** can be located on the ship/rig **230**. In some examples, the system **300** can be connected to a power distribution center **250** on land and/or the system **300** can be designed to be a re-fueling, electric energy station for ocean going electric ships and vessels. The system **300** can be deployed along any coast and/or

along ocean shipping routes as a re-fueling, electric energy station for ocean vessels, ships, aircraft and drones.

[0096] FIG. 4 illustrates another alternate embodiment **400** of an example power system, according to the present disclosure. In the illustrated example, high pressure water can enter the sphere enclosure **110** through intake valve(s) **112a** and rotate the turbine generator **114** to produce electricity. The electricity can then be sent to land and building(s) for distribution to the electrical grid by way of one or more electrical cable(s)/submarine cable(s) **252**.

[0097] High pressure water can exit the sphere enclosure **110** through exit valve(s) **112c**, it can enter tube(s)/pipe(s) **118** delivering high pressure water to water storage tank(s) **120**. Water can then be pumped from the water storage tank **120** to the power distribution center **250** and exit into the ocean. The electrical energy can be delivered from the turbine **114** to a power distribution center **250** or electrical grid **240**.

[0098] The sphere enclosure **110** can have several options as a build, including, but not limited to, mounted to a seabed, including high pressure air tubes required to operate as a submarine, ballast tank(s), variable ballast tank(s), trim tank(s), vent valve(s), pressure hull and required components to operate as a remotely piloted submarine. It can surface up or descend down to certain sea levels.

[0099] The system **400** can be deployed at ocean depths above the seabed or on the seabed. The enclosure **110** can be tethered to land and building(s), such as power distribution centers **250**, in order to deliver electrical energy to cities.

[0100] The system **400** can be designed to produce electricity near the shore in deep waters. It can be utilized as a long term energy platform. It can be deployed as a part of an electricity producing farm at seashore. Main components can be placed in the power distribution center located on land.

[0101] FIG. 5 illustrates another alternate embodiment **500** of an example power system, according to the present disclosure. In the illustrated example of FIG. 5, the sphere enclosure **110** can be tethered to the vessel **230** as a power source while the vessel is stationary or traveling.

[0102] The system **500** comprises a sphere enclosure **110**, pressure valve(s), computer(s), motor(s), tube(s) **118**, cable(s) and/or submarine cable(s), electronics, tether apparatus, and sensor(s). In some examples, the system **500** can produce electrical energy to be stored on vessels and used as a power source.

[0103] In some examples, system **500** can be used to produce electricity for ocean liners, cargo ships, aircraft, drones and ocean vessels. In other examples, it can be used as a mobile power producing source for cities, and other entities that require emergency electrical power, short term. Operation staff employees can work and live aboard the ship or rig **230**. Ship/rig **230** can be deployed anywhere there are deep waters. When system **500** is not connected to a power distribution center on land, it can be designed to be a re-fueling, electric energy station for ocean going ships, aircraft, drones and vessels. In some examples, system **500** can be deployed along coasts and/or along ocean shipping routes as a re-fueling, electric energy station.

[0104] FIG. 6 illustrates another alternate embodiment **600** of an example power system, according to the present disclosure. In the illustrated example of FIG. 6, high pressure water can enter the sphere enclosure **110** through intake valve(s) **112a**. High pressure water can be sent to a power distribution center(s) **250**, turbine generator **114**, valve(s) and/or gear boxes, power transmission, computer and/or electronics required to operate the system **600**. The power distribution center **250** can be located on land, and used to distribute energy to the electrical grid by way of electrical cable(s) and/or submarine cable(s). High pressure water can exit the sphere enclosure **110** through exit valve(s) **112c**, it can enter a tube(s)/pipe(s) **118** delivering high pressure water to power distribution center **250**/electrical grid **240**.

[0105] The enclosure **110** can be deployed at ocean depths above the seabed or on the seabed. The enclosure **110** can be tethered to land and building(s), or power distribution centers **250** in order to deliver electrical energy to an electrical grid. As shown, many of the components of the system **600** can be placed in the power distribution center **250** located on land, and thus no ship or rig is

necessary. The enclosure **110** can be deployed under the sea, near the seashore, or in deep waters. The turbine **114** can be located in the power distribution center **250** located on land.

[0106] FIG. 7 illustrates another alternate embodiment **700** of an example power system, according to the present disclosure. System **700** includes two enclosures **110a**, **110b** (collectively referred to herein as enclosures **110**).

[0107] System **700** can optionally include a tube **220** supplying high air PSI pressure from the vessel **230** to the two sphere enclosures **110**. The air supply can be used to adjust the water pressure in the two enclosures **110**. High pressure water can enter the sphere enclosures **110** through intake valve(s) and exits of the enclosures **110**, on its way to the turbine generator **114** located on the vessel **230**.

[0108] System **700**, as illustrated in FIG. 7, can be mounted to an electrical mining platform rig **230** by way of attachment to a vertical structure and operated up and down as a type of elevator for the sphere enclosures **110** to achieve the required depth and PSI pressure. There can be one or more sphere enclosures **110** attached to the electrical mining platform rig **230**.

[0109] System **700** can be deployed in ocean(s), sea(s), river(s), lake(s), manmade lakes(s) and body of waters, and can operate in water depths and land holes filled with water at depths. It can employ PSI pressure at ocean depths.

[0110] System **700** can have an option to use a mechanical mechanism to move the sphere enclosure up to the sea surface and down to the seabed or use motors deployed to move the sphere enclosure up or down while attached to a main tower.

[0111] System **700** can be used with oil platforms that may need to be modified. Alternatively, a new electrical mining platform rig **230** can be built from scratch. One or many sphere enclosures **110** can be used system **700**. One, two, three, four, five, six or more sphere enclosures can be used to balance and deploy an electrical energy farm at sea.

[0112] In some examples, system **700** can be designed to be deployed on modified oil platforms and/or new electrical mining platform rig(s) designed and built from scratch. The main structure can function like an elevator for the sphere enclosures **110**. It can move the enclosures up for maintenance and operations. It can move them down for deployment at the required depths. The structure can be designed to support and service more than one attached sphere enclosures. This system **700** can be designed to be a fueling/electric energy station at sea, for electric ships, aircraft, drones and vessels. It can be deployed across the coastal areas and across the ocean, near shipping lanes. It can be connected to a power distribution center on land and or designed to be a re-fueling, electric energy station for ocean going electric ships aircraft, drones and vessels. For example, system **700** can be deployed along the world's coast and along ocean shipping routes as re-fueling, electric energy stations. An optional turbine generator can be placed inside the sphere enclosure. It can supply electric power to ocean cities and outpost.

[0113] FIG. 8 illustrates another alternate embodiment **800** of an example power system, according to the present disclosure. System **800** is similar to system **700**. However, system **800** includes turbine generators **114a** and **114b** inside the enclosures **110a** and **110b**. Also, system **800** uses the storage tank **120**.

[0114] FIG. 9 illustrates another alternate embodiment **900** of an example power system, according to the present disclosure. FIG. 9 shows an embodiment of an example power system deployed inside a submarine to power the submarine **910**.

[0115] FIG. 10 illustrates another alternate embodiment **1000** of an example power system, according to the present disclosure. As shown, system **1000** includes the water pump **122** disposed inside the water storage tank **120**.

[0116] System **1000** comprises a sphere enclosure or any shape enclosure that houses the embodiment (device) and does not allow high pressure water to escape, except through intake(s) or outlet(s) valve(s) and/or nozzle(s) that control high pressure water flow. The sphere enclosure includes a turbine generator, valve(s) and/or nozzle(s), gear box, power transmission, computer,

and electronics required to operate system **1000**. In some embodiments, system **1000** further comprises a PSI water tube, a water storage tank, and a water turbo pump.

[0117] In some embodiments, the water turbo pump can be placed inside the water storage tank, near the water storage tank, or outside the water storage tank. In other embodiments, the water turbo pump can discharge water into the sea depths at a higher PSI pressure level than the sea depth PSI pressure level deployed. System **1000** can use pressurized air and gas to move water through the tubes into a sphere enclosure. This can result in extreme PSI pressure that propels a turbine generator and/or one or more turbine generator blade(s), creating continual electrical energy for its power storage or distribution center.

[0118] In other embodiments, in addition to pressurized air and gas, pressurized water is also used to move water through the tubes into a sphere enclosure. This can result in extreme PSI pressure that propels a turbine generator, creating continual electrical energy for its power storage or distribution center. The entire energy system can receive operational power from its turbine generator(s), power storage unit(s), battery, and/or batteries, a combination thereof or the like.

[0119] High pressure water can enter the sphere enclosure through intake valve(s) and/or nozzle(s) and rotate the turbine generator and/or one or more turbine generator blade(s) to produce electricity. The electricity can then be sent to land and building(s) for distribution to the electrical grid by way of one or more electrical cable(s) and/or submarine cable(s).

[0120] The sphere enclosure can include stop valve(s) to be used as emergency turn on/off lever(s) controlling high pressure water flow. The electricity can be produced by hydrokinetic energy obtained from water pressure at ocean depths.

[0121] High pressure water can exit the sphere enclosure through exit valve(s) and/or nozzle(s), it can enter tube(s) and or pipe(s) delivering high pressure water to water storage tank(s). Water can then be pumped from the water storage tank to the power distribution center and exit into the ocean. The electrical energy can be delivered from the turbine to a power distribution center electrical grid.

[0122] FIG. **11** illustrates another alternate embodiment **1100** of an example power system, according to the present disclosure. In the illustrated example, system **1100** comprises a water turbo pump **1110** disposed inside an intake port of the enclosure **110**. In other embodiments, the water turbo pump **1110** can be placed inside the sphere enclosure **110**, at an outlet/output valve(s) and/or nozzle(s) **112c** of the sphere enclosure **110**, near an outlet/output valve(s) and/or nozzle(s) **112c** of the sphere enclosure **110**, or outside the sphere enclosure **110**. In some embodiments, there is a water turbo pump **1110** in place of an outlet/output valve and/or nozzle **112c**. There can be one or more water turbo pump(s) **1110** employed in system **1100**. The water turbo pump **1110** can discharge water into the sea depths at a higher PSI pressure level than the sea depth. The sphere enclosure can include stop valve(s) to be used as emergency turn on/off lever(s) controlling high pressure water flow. The electricity can be produced by hydrokinetic energy obtained from water PSI pressure at ocean depths.

[0123] FIG. **12** illustrates an example embodiment **1200** of an example power system, according to the present disclosure. Instead of utilizing ocean depth pressure, system **1200** can comprise high pressurized water **1202**, pressurized air tank(s) **1220**, and/or water turbo pump(s) **1210** to power the enclosure **110**.

[0124] System **1200** can utilize pressurized air and/or gas, along with pressurized water **1202**, to move water through tubes into a sphere enclosure **110**, resulting in extreme PSI pressure that can propel a turbine generator **114**, creating continual electrical energy for its power storage or distribution center. In some embodiments, the turbine generator includes one or more blade(s) which are rotated to propel the turbine generator **114** and create energy/electricity. Discharged water can be routed to a non-pressurized water tank **1204**. The water is then returned to the pressurized air and water tank **1202**. The water routing process can then start over. The entire energy system can receive operational power from its turbine generator(s) **114**, power storage

unit(s)/batterie(s) **1230**, or combination thereof or the like. In some embodiments, system **1200** can be completely perpetual, except for maintenance and repairs. System **1200** can be deployed nearly anywhere on land or island(s). Land based deployment can result in less government regulations which could be an advantage to utilizing the systems described herein.

[0125] In other embodiments, system **1200** can use water, air, gas, and/or a combination therefore. It can further comprise a sealed plate, floating roof or a bladder to receive the air and/or gas, covering the water. The plate and/or bladder can be pushed down onto the water using PSI pressure, hydraulics, and/or pneumatics systems to create very high PSI pressure. There is no limit on creating PSI pressure with the example embodiments described herein. This can allow the pressurized water to enter the sphere enclosure intake valve(s) and/or nozzle(s) and/or water turbo pump **1210** on its way to rotate the turbine generator(s) **114** to produce electricity. In some embodiments, the turbine generator(s) **114** are rotated via blade(s) to produce electricity, e.g. the turbine generator **114** can include one or more blade(s). This setup can include one, two, or more water tanks. Some embodiments, can include pressurized water tank **1202** and non-pressurized water tank **1204**. In other embodiments, before pressurized water tank **1202** is empty, water tank **1204** can start service to replenish spent water from pressurized tank **1202**. Meanwhile, tank **1204** can start to deliver water into the near empty pressurized water tank **1202**. The cycle repeats itself for periods of time. Time can be in minutes, hours, days, etc. In some embodiments, the time depends on the volume of the tanks **1202** and **1204**. These systems can deliver high PSI pressure to turbine generators **114**.

[0126] Advantages of the systems/example embodiments described herein, include a land based power plant, reusable circulating water, an entire power plant being powered by one of the turbine generators, an energy storage unit being used to start a power plant, power plant construction has less regulations and can be deployed anywhere on land, energy system can be completely perpetual, except for maintenance and repairs, not affected by cold, hot or other weather extremes, can be used to provide broadband internet service to remote locations, and no carbon footprint is left behind.

[0127] FIG. **13** illustrates an alternative embodiment **1300** of an example power system, in accordance with the present disclosure. In some embodiments, system **1300** is deployed anywhere on land, i.e. the ocean is not required. It can include one or more water tank(s) **1310**, **1320**, **1330** inside a renewable energy power plant. In other embodiments, instead of ocean depth pressure, the example embodiment can use high pressurized water, one or more pressurized air tank(s) **1340**, **1342**, and/or one or more water turbo pump(s) **1350**, **1352** to power the embodiment.

[0128] System **1300** can utilize pressurized air and/or gas, along with pressurized water, to move water through tubes into a sphere enclosure, resulting in extreme PSI pressure that can propel a turbine generator, creating continual electrical energy for its power storage or distribution center. In some embodiments, the turbine generator(s) **114** is rotated via blade(s) to produce electricity, e.g. the turbine generator **114** can include one or more blade(s). Discharged water can be routed to a non-pressurized water tank, e.g. water tank **1310**. The water is then returned to the pressurized air and water tank **1330**. The water routing process can then start over. The entire energy system can receive operational power from its turbine generator(s) **114**, power storage unit(s), battery, batteries, and/or a combination thereof, or the like. System **1300** can be deployed nearly anywhere on land or island(s). Land based deployment can result in less government regulations which could be an advantage to utilizing the systems described herein.

[0129] In other embodiments, system **1300** can use water, air, gas, and/or a combination thereof. It can further comprise a sealed plate, floating roof or a bladder to receive the air and/or gas, covering the water. The plate and/or bladder can be pushed down onto the water using PSI pressure, hydraulics, and/or pneumatics systems to create very high PSI pressure. There is no limit on creating PSI pressure with the example embodiments described herein. This can allow the pressurized water to enter the sphere enclosure intake valve and/or nozzle(s) and/or water turbo

pump on its way to rotate the turbine generator(s) to produce electricity. In some embodiments, the turbine generator(s) are rotated via blade(s) to produce electricity, e.g. the turbine generator can include one or more blade(s). This setup can include one, two, or more water tanks. Some embodiments, can include water tanks **1320**, **1330**, and **1310**. In other embodiments, before water tank **1310** is empty, water tank **1330** can start service and take the place of water tank **1310** by delivering water into the sphere enclosure intake valve(s) and/or nozzle(s) and/or water turbo pump on its way to rotate the turbine generator(s) **114** to produce electricity. In some embodiments, the turbine generator(s) are rotated via blade(s) to produce electricity, e.g. the turbine generator can include one or more blade(s). Water discharged from the sphere enclosure can be routed to water tank **1310**. In some embodiments, if needed water in water tank **1310** can send the discharged water to replenish spent water from tank **1320** and tank **1330** through a water turbo pump. In some embodiments, water tank **1320**, water tank **1330**, and/or water tank **1310** can be pressurized. Meanwhile, tank **1330** can start to deliver water into the near empty pressurized water tank **1320**. The cycle can repeat itself for periods of time. Time can be in minutes, hours, days, etc. In some embodiments, the time depends on the volume of the tanks. These systems can deliver high PSI pressure to turbine generators.

[0130] Advantages of the systems/example embodiments described herein, include a land based power plant, reusable circulating water, an entire power plant being powered by one of the turbine generators, an energy storage unit being used to start a power plant, power plant construction has less regulations and can be deployed anywhere on land, energy system can be completely perpetual, except for maintenance and repairs, it is not affected by cold, hot or other weather extremes, it can provide broadband internet service to remote locations, and leaves no carbon footprint behind.

[0131] In some examples, the sphere enclosure ocean PSI pressure at intake valve and/or nozzle can employ Penstock Pipe **120** in (10 ft) Diameter.

[0132] The preceding disclosures are illustrative embodiments. It should be appreciated by those of skill in the art that the devices, techniques and methods disclosed herein elucidate representative embodiments that function well in the practice of the present disclosure. However, those of skill in the art should, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments that are disclosed and still obtain a like or similar result without departing from the spirit and scope of the invention.

[0133] Unless otherwise indicated, all numbers expressing quantities of ingredients, properties such as molecular weight, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present invention. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

[0134] The terms “a” and “an” and “the” and similar referents used in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context.

Recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein

or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. “such as”) provided herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

[0135] The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.”

[0136] Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member may be referred to and claimed individually or in any combination with other members of the group or other elements found herein. It is anticipated that one or more members of a group may be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

[0137] Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Of course, variations on those preferred embodiments will become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventor expects those of ordinary skill in the art to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

[0138] Specific embodiments disclosed herein may be further limited in the claims using consisting of or consisting essentially of language. When used in the claims, whether as filed or added per amendment, the transition term “consisting of” excludes any element, step, or ingredient not specified in the claims. The transition term “consisting essentially of” limits the scope of a claim to the specified materials or steps and those that do not materially affect the basic and novel characteristic(s). Embodiments of the invention so claimed are inherently or expressly described and enabled herein.

[0139] Further, it is to be understood that the embodiments of the invention disclosed herein are illustrative of the principles of the present invention. Other modifications that may be employed are within the scope of the invention. Thus, by way of example, but not of limitation, alternative configurations of the present invention may be utilized in accordance with the teachings herein. Accordingly, the present invention is not limited to that precisely as shown and described.

Claims

1. A system comprising: at least two enclosures submerged in a body of water; at least two ports, each having at least one intake valve and at least one exit valve, configured to be disposed along a periphery of the at least two enclosures; and a turbine generator located on a vessel; wherein high-pressure water enters and exits the at least two enclosures through the at least two ports in transport to the turbine generator.
2. The system of claim 1, further comprising a tube configured to supply high air pounds per square inch (PSI) pressure from an air compressor located on the vessel to the at least two enclosures.
3. The system of claim 2, wherein the supplied high air PSI pressure is configured to adjust water pressure in the at least two enclosures.
4. The system of claim 3, wherein the high-pressure water exits the at least two enclosures and is transported to the turbine generators via one or more water tubes.

5. The system of claim 4, wherein the high-pressure water rotates the turbine generators to produce electricity on the vessel.
 6. The system of claim 5, wherein the vessel is an offshore platform.
 7. The system of claim 5, wherein the vessel is a fueling/electrical energy station for electric ships, aircrafts, drones, vessels, and the like.
 8. The system of claim 5, wherein the system is mounted to an electrical mining platform rig by way of attachment to a vertical structure.
 9. The system of 8, wherein the system is operated vertically in a raising and lowering motion for the at least two enclosures to achieve a required depth and PSI pressure.
 10. The system of claim 9, wherein the at least two enclosures are raised up to a sea surface for maintenance and operations.
 11. The system of claim 9, wherein the at least two enclosures are lowered down to a seabed for deployment at required depths.
 12. The system of 9, wherein the system is connected to a power distribution center on land configured to deliver the electricity to ocean cities and outposts.
 13. The system of claim 9, optionally further comprising an additional turbine generator in at least one of the at least two enclosures.
 14. The system of claim 1, further comprising one or more additional enclosures.
 15. A system comprising: at least two enclosures, each comprising a turbine generator, submerged in a body of water; at least two ports, each having at least one intake valve and at least one exit valve, configured to be disposed along a periphery of the at least two enclosures; and wherein high-pressure water enters the at least two enclosures through the at least two ports.
 16. The system of claim 15, further comprising a tube configured to supply high air PSI pressure from an air compressor located on a vessel to the at least two enclosures.
 17. The system of claim 16, wherein the supplied high air PSI pressure is configured to adjust water pressure in the at least two enclosures.
 18. The system of claim 17, wherein the high-pressure water entering the at least two enclosures rotates the turbine generators to produce electricity.
 19. The system of claim 18, wherein the electricity is sent to the vessel for distribution by way of one or more electrical cables.
 20. The system of 18, optionally further comprising a mechanical mechanism configured to move the at least two enclosures up to a sea surface and/or down to a seabed.
 21. The system of claim 18, further comprising a water storage tank coupled to the turbine generators to receive the water flowing out of the turbine generators.
 22. The system of claim 21, wherein the air compressor is configured to deliver high air PSI pressure to the water storage tank.
 23. The system of claim 22, wherein the water discharged from the water storage tank is transported to at least one water pump located on the vessel by way of one or more water tubes.
 24. The system of claim 15, further comprising one or more additional enclosures.
 25. The system of claim 23, wherein the vessel is an electrical refueling station platform rig.
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